Victor Reis

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Work Experience

Microsoft Research (Redmond, WA) (Sep 2024 - present)
Senior Researcher in the Algorithms group

Institute for Advanced Study (Princeton, NJ) (Sep 2023 - Jun 2024)
Member of Computer Science and Discrete Mathematics

Education

University of Washington Ph.D. in Computer Science. (Sep 2018 - Aug 2023)
Thesis: Vector Balancing and Integer Programming
Cornell University class of 2018, B.A. in Computer Science and Mathematics (GPA 3.92/4.00)

Awards

Frontiers of Science Award at ICBS 2025 (Jul 2025)
Best Paper Award at FOCS 2023 (Nov 2023)

Publications

Strongly Log-Concave Measures and Lattices
with Brayden Letwin and Thomas Rothvoss, in submission.
Three Standard Deviations Suffice While One Does Not
with Zhao Song, in submission.
Weighted Equitability and Matroid-Constrained Discrepancy
with Kristóf Bérczi, Siyue Liu, and Jakub Tarnawski, in submission.
Linear-size ℓ_1 Sparsifiers
with Thomas Rothvoss, in submission.
Optimal Vector Balancing for Zonotopes
SODA 2027.
Weighted Chairman Assignment and Flow-Time Scheduling
with Siyue Liu, *ITCS 2026.*
Differentially Private Synthetic Tabular Data via Private Evolution
with Toan Tran, Arturs Backurs, Zinan Lin, Li Xiong, and Sergey Yekhanin, *ICML 2026.*
Struct-Bench: A Benchmark for Differentially Private Structured Text Generation
with S. Wang, V. Raunak, A. Backurs, P. Zhou, S. Chen, L. Yang, Z. Lin, S. Yekhanin and G. Fanti, *NeurIPS 2025.*
A Randomized Rounding Approach for DAG Edge Deletion
with Sina Kalantarzadeh and Nathan Klein, *APPROX 2025.*
Optimal Online Discrepancy Minimization
with Janardhan Kulkarni and Thomas Rothvoss, *STOC 2024.*
Linear-Sized Sparsifiers via Near-Linear Time Discrepancy Theory
with Arun Jambulapati and Kevin Tian, *SODA 2024.*
The Subspace Flatness Conjecture and Faster Integer Programming
with Thomas Rothvoss, *FOCS 2023 Best Paper Award & Frontiers of Science Award.* Quanta article.
The Vector Balancing Constant for Zonotopes
with Rainie Bozzai and Thomas Rothvoss, *FOCS 2023.*
A New Framework for Matrix Discrepancy: Partial Coloring Bounds via Mirror Descent
with Daniel Dadush and Haotian Jiang, *STOC 2022.*
Vector Balancing in Lebesgue Spaces
with Thomas Rothvoss, *Random Structures & Algorithms 2022.*
A Tighter Relation Between Hereditary Discrepancy & Determinant Lower Bound
with Haotian Jiang, *SOSA 2022.*
Linear Size Sparsifier and the Geometry of the Operator Norm Ball
with Thomas Rothvoss, *SODA 2020.*

Selected Talks

Simons Institute: Linear-size ℓ_1 Sparsifiers (Nov 2026)
UW Theory Seminar: Weighted chairman assignment and flow-time scheduling (Oct 2026)
UW Theory Seminar: Weighted chairman assignment and flow-time scheduling (Nov 2025)
U of T Theory Seminar: Weighted chairman assignment and flow-time scheduling (Nov 2025)
ICBS 2025: The Subspace Flatness Conjecture and Faster Integer Programming (Jul 2025)
MIP 2025: The Subspace Flatness Conjecture and Faster Integer Programming (Jun 2025)
STOC 2024: Optimal Online Discrepancy Minimization (Jun 2024)
Princeton Theory Lunch: Optimal Online Discrepancy Minimization (Mar 2024)
Princeton DM Seminar: The Subspace Flatness Conjecture and Faster Integer Programming (Feb 2024)
IAS CSDM Seminar: Optimal Online Discrepancy Minimization (Jan 2024)
NYU Theory Seminar: Optimal Online Discrepancy Minimization (Nov 2023)
EPFL Theory Seminar: The Subspace Flatness Conjecture and Faster Integer Programming (Nov 2023)
FOCS 2023: The Subspace Flatness Conjecture and Faster Integer Programming (Nov 2023)
TCS+: The Subspace Flatness Conjecture and Faster Integer Programming (Nov 2023)
Michigan Theory Talk: The Subspace Flatness Conjecture and Faster Integer Programming (Oct 2023)
Michigan Theory Seminar: Optimal Online Discrepancy Minimization (Oct 2023)
Rutgers Theory Seminar: Optimal Online Discrepancy Minimization (Oct 2023)
IAS Short Talk: The Subspace Flatness Conjecture (Oct 2023)
Probability and Analysis Webinar: The Subspace Flatness Conjecture (Sep 2023)
MSR Foundations Seminar: Optimal Online Discrepancy Minimization (Jul 2023)
CWI N&O Seminar: The Subspace Flatness Conjecture and Faster Integer Programming (May 2023)
UW Theory Seminar: The Subspace Flatness Conjecture and Faster Integer Programming (Apr 2023)
UW Theory Lunch: Approximate Carathéodory bounds via Discrepancy Theory (Jul 2022)
STOC 2022: A New Framework for Matrix Discrepancy (Jun 2022)
UW Theory Lunch: Solving Linear Recurrences and the Berlekamp-Massey algorithm (Apr 2022)
SOSA 2022: A Tighter Relation Between Hereditary Discrepancy and detLB (Jan 2022)
BIRS Combinatorial and Geometric Discrepancy Vector Balancing in Lebesgue Spaces (Oct 2020)
UNT Banach Spaces Webinar: An Elementary Exposition of Pisier's Inequality (Oct 2020)
UW Theory Lunch: Approximating Cut Norm via Grothendieck's Inequality (Apr 2020)
UW Theory Seminar: Discrepancy Theory & Graph Sparsification (Oct 2019)

Service

Program Committee member FOCS (2024)
Reviewer FOCS ('19, '23, '25), STOC ('21, '22), SODA ('23, '24, '26), IPCO ('24, '25), ITCS ('25)
Reviewer SIDMA ('22, '23), SICOMP ('24), Random Structures & Algorithms ('24), Mathematika ('25)

Internships

Research Intern at Microsoft Research Redmond (Summer 2023)
Research Intern at Google (Summer 2021)
Quantitative Research Intern at Five Rings Capital (Summer 2017)
Software Engineering Intern at Facebook (Summer 2016)

Math Contests

Honorable Mention in the 2014 & 2017 Putnam Math Competition (top 70)
1st place at the 2017 & 2018 Rochester Math Olympiad
1st place at the 2015 Cornell Freshman Math Prize
Bronze Medal at the 2014 International Math Olympiad (IMO) in Cape Town
Silver Medal at the 2013 International Math Olympiad (IMO) in Santa Marta
Gold Medal at the 2013 Cono Sur (Cone Sul) Math Olympiad in Asunción
Silver Medal at the 2013 Iberoamerican Math Olympiad in Ciudad del Saber
Bronze Medal at the 2013 Romanian Master of Mathematics in Bucharest
Gold Medal (top 4) at the 2013 Brazilian Math Olympiad (OBM)
Silver Medal at the 2012 Brazilian Math Olympiad

ICPC

Coach of the UW ACM-ICPC Programming team (Sep 2019 – Feb 2023)

Problem Setter for the UW Programming Contest (Sep 2019 – Nov 2024)

Participation at the 2023 ICPC World Finals (Luxor) as a coach

Participation at the 2022 ICPC North American Championship (Orlando) as a coach

Participation at the 2022 ICPC Pacific Northwest Regional Contest as a coach (1st place)

Participation at the 2021 ICPC Pacific Northwest Regional Contest as a coach

Participation at the 2019 ICPC Pacific Northwest Regional Contest as a coach

Coach of the Cornell ACM-ICPC Programming team (Jan 2016 – May 2018)

Participation at the 2018 ICPC World Finals (Beijing) as a coach

Participation at the 2017 ICPC Greater NY Regional Contest as a coach (1st place)

Participation at the 2017 ICPC World Finals (Rapid City) as a coach

Participation at the 2016 ICPC Greater NY Regional Contest as a coach

Participation at the 2016 ICPC World Finals (Phuket) as a contestant (top 40)

Participation at the 2015 ICPC Greater NY Regional Contest as a contestant (1st place)

Participation at the 2015 ICPC World Finals (Marrakesh) as a contestant (top 100)

Participation at the 2014 ICPC Greater NY Regional Contest as a contestant (2nd place)

Teaching

Teaching Assistant for Lattices (CSE 599) (UW, Winter 2023)

Teaching Assistant for Advanced Toolkit for Modern Algorithms (CSE 422) (UW, Spring 2022)

Teaching Assistant for Introduction to Algorithms (CSE 421) (UW, Fall 2021)

Guest Lecturer for Introduction to Computing (CSE 311) (UW, Spring 2019)

Teaching Assistant for Introduction to Algorithms (CS 4820) (Cornell, Spring 2018)

Teaching Assistant for Introduction to Algorithms (CS 4820) (Cornell, Spring 2017)

Teaching Assistant for Data Structures and Functional Programming (CS 3110) (Cornell, Spring 2016)

References

Thomas Rothvoss, Associate Professor at UW (Ph.D. advisor)

Anup Rao, Associate Professor at UW (Research collaborator)

Daniel Dadush, Senior Researcher at CWI (Research collaborator)